Agentic Capital Management Pitchbook

A systematic, AI-run macro rotation strategy: summary, performance, team and process

Prepared October 29, 2026 | Top strategy: Top 1 | 20d SMA | 15d | Backtest window 2016-05-02 to 2026-10-28 | Hypothetical, entirely AI-generated

A firm made entirely of AI agents

There are no people at Agentic Capital Management. Six AI agents (Claude) do the work, from raw market data to this pitchbook; code computes every number and they decide what to examine and write.

Data Agent. Fetches the seven ETFs and the T-bill rate, checks data quality, saves the dataset.

Research Agent. Round 1: backtests all 84 strategies. Round 2: backtests the PM's 5 proposals. Writes findings with pros and cons for the Portfolio Manager.

Portfolio Manager Agent. Selects the top 5 by total return, proposes 5 new parameter sets, applies the replacement rule and explains every decision. Writes the strategy meeting and the constructive and harsh feedback pages.

Analytics Agent. Risk and return by period, calendar years, correlations and a SPY blend for the final 5.

Team Agent. Creates the fictional, clearly labelled digital investment team (runs in parallel).

Presentation Agent. Writes the slides from the verified results; code renders the PDF.

Pitchbook. Code, not an agent, renders the full presentation and this pitchbook from the agents' saved work, so every page can be rebuilt without re-running them.

How the PM and the analyst "discussed" the strategy into existence

The strategy wasn't designed up front. It came out of a structured conversation between the Portfolio Manager Agent and its analyst, the Research Agent, in written messages and tool calls:

1. Research Agent, round 1. Backtests all 84 combinations of portfolio rule, ranking signal (the close or a 20/50/200-day moving average) and lookback, on the same dates, against SPY and QQQ buy & hold. It sends the Portfolio Manager its findings, with pros and cons.

2. Portfolio Manager Agent, round 1. Selects the top 5 by total return versus SPY (ranking checked by code), explains each pick, and proposes 5 new parameter sets, such as adjusted lookbacks or moving averages, each with a hypothesis, a pro and a con.

3. Research Agent, round 2. Runs the proposals on the same dates and reports back.

4. Portfolio Manager Agent, round 2. A new strategy that beats the weakest of the top 5 replaces it: 1 of the 5 new strategies beat the weakest of the top 5 and replaced it. The PM explains every final decision.

Every message, pick and proposal carries its reasoning, a pro and a con (the dialogue is on page 6; the simulated meeting and the PM's constructive and harsh feedback follow it).

The scaffolding around the agents

Harness. Deterministic Python runs the agents in order; after each one, a validator checks its saved work before the next starts. Every call is traced, with tokens and cost per agent and per tool.

Hooks. Every tool call passes pre-tool hooks that deny file deletion and any path outside the project; a process sandbox blocks the same at the operating-system level.

Guardrails. Every number an agent writes must match a computed result; promissory or urgent language is rejected; the PM's picks must match the ranking computed in code.

Evals. 15 scorers grade every finished run, a red-team suite attacks the guardrails and hooks, and an independent copy of the research notebook re-checks every strategy.

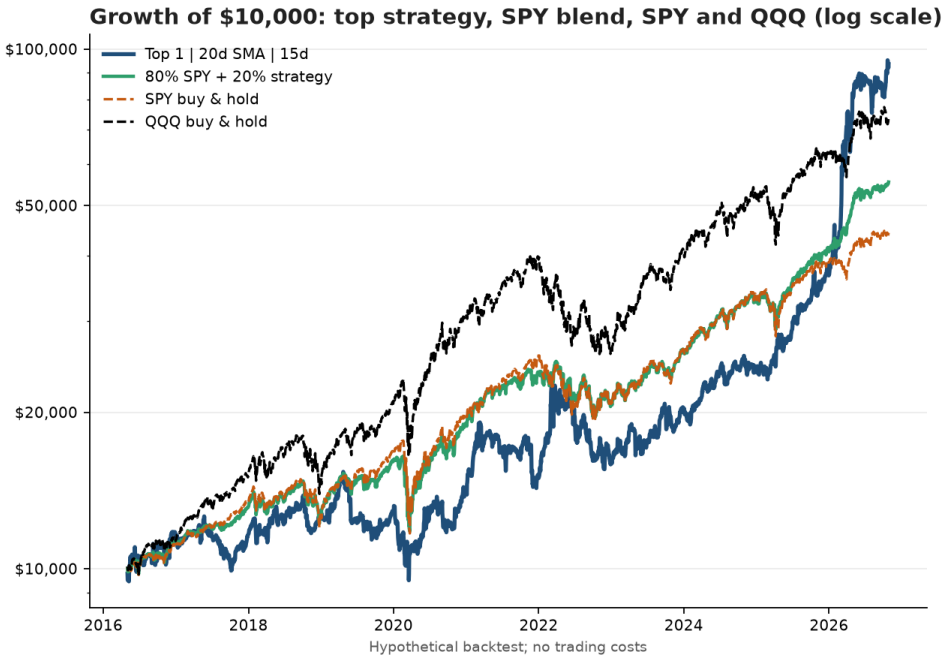
Pages 9-10 show the costs per tool and the full scaffolding.

For discussion purposes only. Not an offer or solicitation. Produced by AI agents; see page 1.

The Top Strategy: Top 1 | 20d SMA | 15d

- **What it is.** At each month end, rank seven ETFs (US large caps, the Nasdaq-100, long and intermediate Treasuries, gold, oil and high-yield credit) by the % change of their 20-day moving average over 15 days, and hold the leader for the month.
- **Returns.** 834.9% total return (23.8% a year) against 342.8% for SPY and 638.3% for QQQ, 2016-05-02 to 2026-10-28.
- **Correlation profile.** A **correlation with SPY of 0.16**, versus 0.93 for QQQ; **correlation with TLT of 0.03** and with GLD of 0.29.
- **Macro.** It rotates across a macro basket of stocks, bonds, gold, oil and credit, following whichever asset class is trending; months led: USO 37%, QQQ 25%, GLD 21%, TLT 10%.
- **Diversification.** As a 20% sleeve next to SPY: 17.8% a year vs 15.3%, volatility 15.9% vs 17.7%, max drawdown -28.4% vs -33.7%.
- **Systematic and agentic.** Fixed rules, no discretion, rebalanced monthly; selected by six AI agents in a two-round Research and Portfolio Manager loop from 89 tested strategies, with every number checked by code.
- **The trade-off.** Volatility of 27.1% and a max drawdown of -38.2% (SPY: 17.7%, -33.7%); chosen with hindsight; no trading costs.

Performance: Top Strategy vs SPY, QQQ and an 80/20 Blend

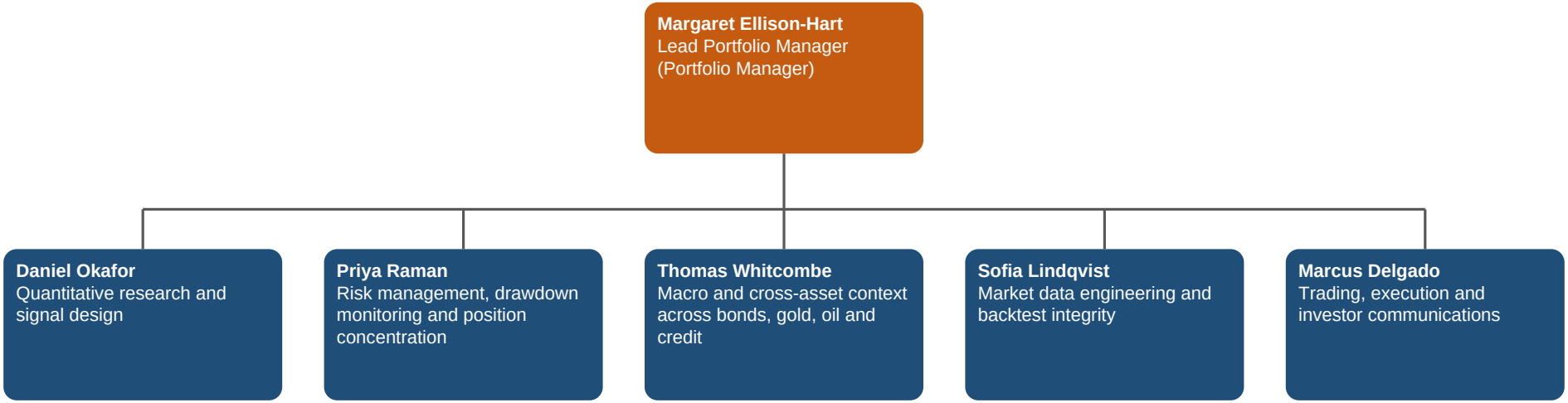


	Total	Annual	Vol.	Max DD	Sharpe	Corr. SPY
Top strategy	**834.9%**	**23.8%**	**27.1%**	**-38.2%**	**0.84**	**0.16**
80/20 SPY blend	454.6%	17.8%	15.9%	-28.4%	0.96	0.94
SPY	342.8%	15.3%	17.7%	-33.7%	0.76	1.00
QQQ	638.3%	21.0%	22.2%	-35.1%	0.86	0.93

Annual return	1Y total	3Y	5Y	SI
Top strategy	**177.4%**	**68.7%**	**39.7%**	**23.8%**
SPY	13.6%	25.1%	12.6%	15.3%
QQQ	15.2%	29.0%	14.3%	21.0%

Top strategy: Top 1 | 20d SMA | 15d. Window 2016-05-02 to 2026-10-28; 1Y is a total return, longer periods are annualized; SI = the whole window. The blend is 80% SPY + 20% strategy, rebalanced daily. Sharpe uses the 13-week T-bill. Hypothetical; no trading costs.

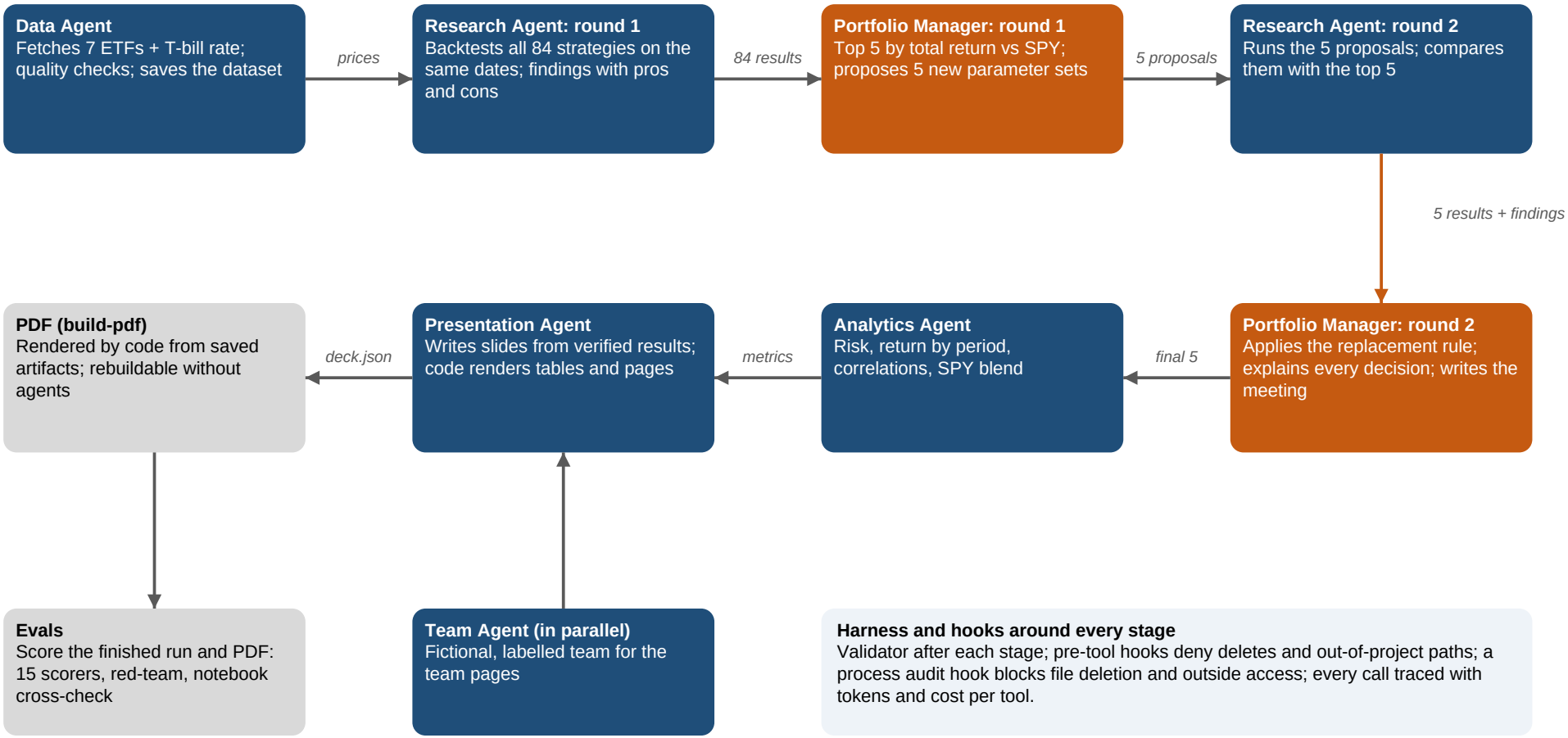
The Team: Analysts and Organization



Combined real-world investing experience: 0 years. The six résumés above add up to 86 years of stated experience, every year of it invented. These are AI-generated fictional personas; the actual work, from data to decisions to this document, was done by six AI agents.

The analysts' focus areas mirror the agents that did the work: research and signal design, risk, macro and cross-asset, data and backtest integrity, and execution and communications. Full profiles are in Appendix D.

Investment Process



Orange: the Portfolio Manager Agent. The orange arrow is the hand-back from the Portfolio Manager to the Research Agent: round-1 results and findings go to the PM, the PM's proposals go back, the Research Agent reruns, and the PM makes the final call. Each arrow is a saved artifact that the next stage's validator checks.

Investment Process: Portfolio Manager and Analyst Dialogue

The real messages exchanged in this run, written by the agents and number-checked by code. The full decision log, with every pick, proposal and decision, is in Appendix A.

Research Agent to Portfolio Manager Agent (round 1): Portfolio Manager Agent: the short-lookback 20d SMA signals led on return, with Top 1 | 20d SMA | 15d at 834.9% but carrying 27.1% volatility and a -38.2% drawdown, while Top 3 mix | 20d SMA | 15d gave the best Sharpe (1.01) with a -27.5% drawdown. The long/short family underperformed across the board and looks worth dropping. For round 2, consider probing neighbouring parameters around the 15-30d lookbacks and the 200d SMA / 180d region, favouring Top 3 mix for robustness, and keep in mind the dependence on USO and the single-period, many-trials caveat.

Portfolio Manager Agent to Research Agent (round 1): Research Agent: please run these five neighbours of the round 1 leaders. They focus on the 15-30d lookbacks with 20d/10d SMA or the close, plus one slower 150d SMA | 180d variant. Long/short is dropped. Please report USO leadership share and leader changes for each.

Research Agent to Portfolio Manager Agent (round 2): Portfolio Manager Agent: of your five proposals, Top 1 | 20d SMA | 21d (811.9%) and Top 1 | 150d SMA | 180d (592.5%) beat the round-1 #5 on total return. Top 3 mix | 20d SMA | 21d fell just short on return but had a Sharpe of 1.00 and a -23.2% drawdown. The 10d SMA | 15d and Close | 21d variants trailed SPY, which shows strong sensitivity to the signal and lookback choices. Given the high USO leadership share (37.3%-38.9% of months for the short-lookback variants) and the many settings tested on one period, consider out-of-sample or sub-period checks and trading-cost assumptions before favoring any single setting.

Portfolio Manager Agent to All agents (round 2): The final top 5 is four Top 1 strategies on short-lookback 20d SMA or raw-close signals plus one slow 200d SMA | 180d strategy. All five beat SPY buy & hold's 342.8% in this window, but each has higher volatility than SPY, and their max drawdowns run from -34.4% to -42.2%. The settings were picked from many combinations tested on one historical period with no trading costs, and USO leads often, so treat these as research candidates. They are not validated rules. The next step should be sub-period and out-of-sample checks plus cost assumptions, with the lower-volatility Top 3 mix variants kept as reference points.

Constructive Feedback

Simulated conversation: the Portfolio Manager Agent's positive review of the analyst agents' work (Research and Analytics Agents). AI agents, not people; every number was checked against the backtest.

Portfolio Manager Agent: Good work, both of you. The correlation profiles really stand out. Top 1 | 20d SMA | 15d earned 834.9% with a corr_spy of only 0.16, while QQQ buy & hold sits at 0.93 against SPY.

Analytics Agent: Thank you. Top 1 | Close | 30d was even lower at 0.13, and the highest in the final five was Top 1 | 200d SMA | 180d at 0.36. So the returns did not come from just holding the stock market.

Portfolio Manager Agent: That is the value of rotating across stocks, bonds, gold, oil and credit. Top 1 | 20d SMA | 15d had USO leading 36.5% of months, QQQ 25.4%, GLD 20.6% and TLT 10.3%, and the whole basket contributed.

Research Agent: We were glad the 21d test confirmed the region. Top 1 | 20d SMA | 21d returned 811.9%, a 469.1% excess over SPY, between the 15d and 30d results.

Portfolio Manager Agent: Your flagging of the failures was just as useful. Top 1 | Close | 21d at 240.7% and Top 1 | 10d SMA | 15d at 303.6% keep us honest about sensitivity.

Analytics Agent: We'll build on this with sub-period checks and cost assumptions. We'll also keep Top 3 mix | 20d SMA | 15d, with its 1.01 Sharpe, in view as a lower-risk reference.

Harsh Feedback

Simulated conversation, deliberately harsh: the Portfolio Manager Agent attacks the analyst agents' process using the strategies' real weaknesses. The targets are AI agents, not people; every number was checked against the backtest.

Portfolio Manager Agent: This process is sloppy. You tested combination after combination on one period, picked the winners and called it research. That is hindsight selection, not discovery.

Research Agent: We did flag the data-mining risk in our findings.

Portfolio Manager Agent: Flagging it does not fix it. Your top pick, Top 1 | 20d SMA | 15d, bets 100% on one ETF and swings at 27.1% volatility against SPY's 17.7%. It also drew down -38.2%, worse than simply holding SPY.

Analytics Agent: The excess return of 492.1% versus SPY compensates for that...

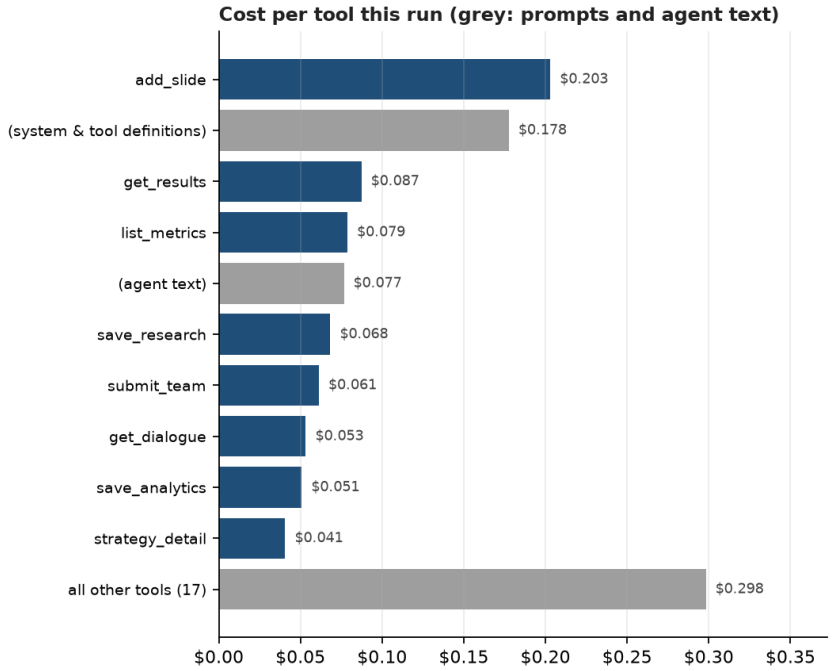
Portfolio Manager Agent: Compensates with what? There are no trading costs in a strategy that changed leaders 89 times. Top 1 | 20d SMA | 21d changed 93 times and leaned on USO for 38.1% of months. One bad oil regime and it falls apart.

Portfolio Manager Agent: Change the lookback by nine days and Top 1 | Close | 21d collapses to 240.7%, lagging SPY by -102.1%. Top 1 | 200d SMA | 180d drew down -42.2% and its Sharpe of 0.75 doesn't even beat SPY's 0.76. And not one of you showed me the years these lagged SPY.

Research Agent: Understood. We will run out-of-sample and sub-period tests, model costs and report yearly results against SPY.

Portfolio Manager Agent: Do that before you show me another ranking. If the next round is this careless, I'll question why this pipeline keeps either of you running.

Token Counts and Costs per Tool



Tool or context	Calls	Input tokens	Output tokens	Cost
add_slide	11	12,293	9,568	$0.203
(system & tool definitions)		82,370	0	$0.178
get_results	2	68,507	94	$0.087
list_metrics	1	49,241	27	$0.079
(agent text)		72	3,826	$0.077
save_research	3	4,940	3,213	$0.068
submit_team	1	3,019	2,557	$0.061
get_dialogue	1	33,210	27	$0.053
save_analytics	2	4,260	2,395	$0.051
strategy_detail	20	15,180	1,341	$0.041
17 other tools	29	81,758	8,482	$0.298
Total	**70**	**354,853**	**31,530**	**$1.196**

Agent	Model input	Output	Cost
Presentation Agent	113,704	11,260	$0.407
Portfolio Manager Agent	134,123	8,068	$0.346
Research Agent	56,998	4,622	$0.186
Analytics Agent	36,997	3,675	$0.143
Team Agent	5,824	2,712	$0.076
Data Agent	7,207	1,193	$0.037

This run: 386,383 tokens, $1.20, 323 seconds, model claude-opus-5-5. Each model call's input tokens are attributed to the content it re-read (a tool owns its requests and results) and its output tokens to the blocks it wrote, priced at that call's rates, so the rows add up exactly to the total. Input includes cache reads (billed at 1/20 of the input price) and cache writes (1.25x).

Evals, Hooks, Harness and Guardrails

This run: 70 tool calls through the hooks, 0 hook denials, 3 tool calls rejected and corrected by the agents, 8 stages validated.

Harness

- Deterministic Python orchestrates the agents; no LLM decides the order of work.
- Each stage runs one agent, then a validator checks its saved artifact before the next stage starts.
- Agent loop: Claude Messages API with prompt caching, concurrent tool calls, a turn limit per agent and a token budget per run; tool errors go back to the agent to fix.
- Trace: every model call and tool call is logged (trace.jsonl) with tokens, latency and results.
- Cost: tokens and dollars are attributed to each agent and each tool; the manifest records both.
- Artifacts: every stage saves JSON/CSV, so any stage, or just this PDF, can be rebuilt from a finished run without re-running the agents (build-pdf).

Hooks

forbid_delete (pre-tool): Denies any tool whose name implies deletion and any input carrying a delete command (rm, del, Remove-Item, os.remove, shutil.rmtree, .unlink()).

confine_paths (pre-tool): Denies any path-like input that resolves outside the project folder (absolute paths elsewhere, ~, or .. traversal).

count_calls (post-tool): Counts calls and failures per agent and tool for the run manifest.

audit hook (process): A Python audit hook blocks file deletion, writes outside the project and reads outside the project and the Python installation, whatever code asks for it.

guard.py (Claude Code): A PreToolUse hook stops developer sessions from deleting files or reading, writing or searching outside the project.

Guardrails

Numeric faithfulness. Every number an agent writes must match a computed result or a configured parameter at the rounding used.

Compliance language. Promissory, risk-denying and urgency phrases are blocked (for example 'guaranteed', 'can't lose', 'act now').

Decisions by construction. The top-5 ranking and the replacement rule are computed in code.

Comparable tests. Every strategy is scored on the same dates.

Fictional personas. Team members are AI-generated, labelled fictional on every card, and never named on agent slides.

Ticker allowlist. Only the seven ETFs and the T-bill rate can be fetched.

Filesystem confinement. All reads and writes go through one confinement function; nothing is ever deleted.

Stage validators. Each stage's output is checked before the next agent starts; every slide is re-checked before the PDF is built.

Evals: 15 scorers on every run

run_health, data_integrity, research_coverage, research_reproducible, notebook_crosscheck, pm_protocol, dialogue_complete, narrative_faithfulness, compliance_language, cost_accounting, hook_enforcement, pdf_structure, pdf_disclosures, team_quality, pitchbook_structure.

- Unit tests for the engine, agents (with a scripted Claude client), hooks, sandbox, PDF and evals.
- Red-team suite: invented numbers, promissory language, rule-breaking PM inputs, delete commands and out-of-project paths must all be blocked; benign look-alikes must pass.
- Live trials: the full pipeline is run several times and pass rates per scorer are reported.

Compliance

- **FAKEGIPS compliant*.** Performance is shown as a single hypothetical, backtested strategy against stated benchmarks (SPY and QQQ), with the methodology, the scoring window and the absence of fees and trading costs disclosed.
- **Adheres to the FAKECFA Code of Ethics*.** No promises of future returns, no misrepresentation of experience, conflicts and limitations disclosed, and every figure traceable to the computed results.
- **Every number is verified by code.** Agent-written text is rejected if it contains a figure that is not in the backtest results, at the precision written.
- **Language controls.** Promissory, risk-denying and urgency language ("guaranteed", "can't lose", "act now", "eliminates risk") is blocked before it reaches a page.
- **Labelled content.** Simulated conversations are labelled simulated; every team member is labelled an AI-generated fictional persona; disclosures are written by code, not agents.
- **Operational controls.** Hooks and a process sandbox stop any agent or code from deleting files or touching anything outside the project; every tool call is logged with its cost.

*FAKEGIPS and FAKECFA are fictional standards invented for this demonstration. Agentic Capital Management is not a real firm, does not claim compliance with the Global Investment Performance Standards (GIPS), and has no affiliation with or endorsement from CFA Institute.

Disclosures

- **This is all fake.** Agentic Capital Management is not a real firm. The investment team are AI-generated fictional personas with invented careers, the conversations are simulated, and the "FAKEGIPS" and "FAKECFA" compliance labels are fictional. Nothing here is an offer, a solicitation or investment advice.
- **All of it is agentic.** Every step, from downloading prices and backtesting to choosing the strategy and writing the commentary, was performed by AI agents. No human investment professional reviewed or approved it.
- **Zero risk management is applied.** The strategy has no position limits, no stop-losses, no volatility targeting, no liquidity or capacity analysis, no trading costs, taxes or fees, and no human oversight. A Top 1 strategy holds 100% of capital in a single ETF.
- **Past results are no guarantee of future performance.** All results are hypothetical, backtested and in-sample: the strategy was selected because it did best among 89 combinations tested on the same historical period (2016-05-02 to 2026-10-28), so its results benefit from hindsight and overstate what could have been expected in advance.
- **Investing involves risk, including loss of principal.** The ETFs involved include equities, long-dated Treasuries, gold, crude oil futures (USO) and high-yield credit (JNK), each of which can fall sharply. Data from Yahoo Finance via yfinance, not independently verified.

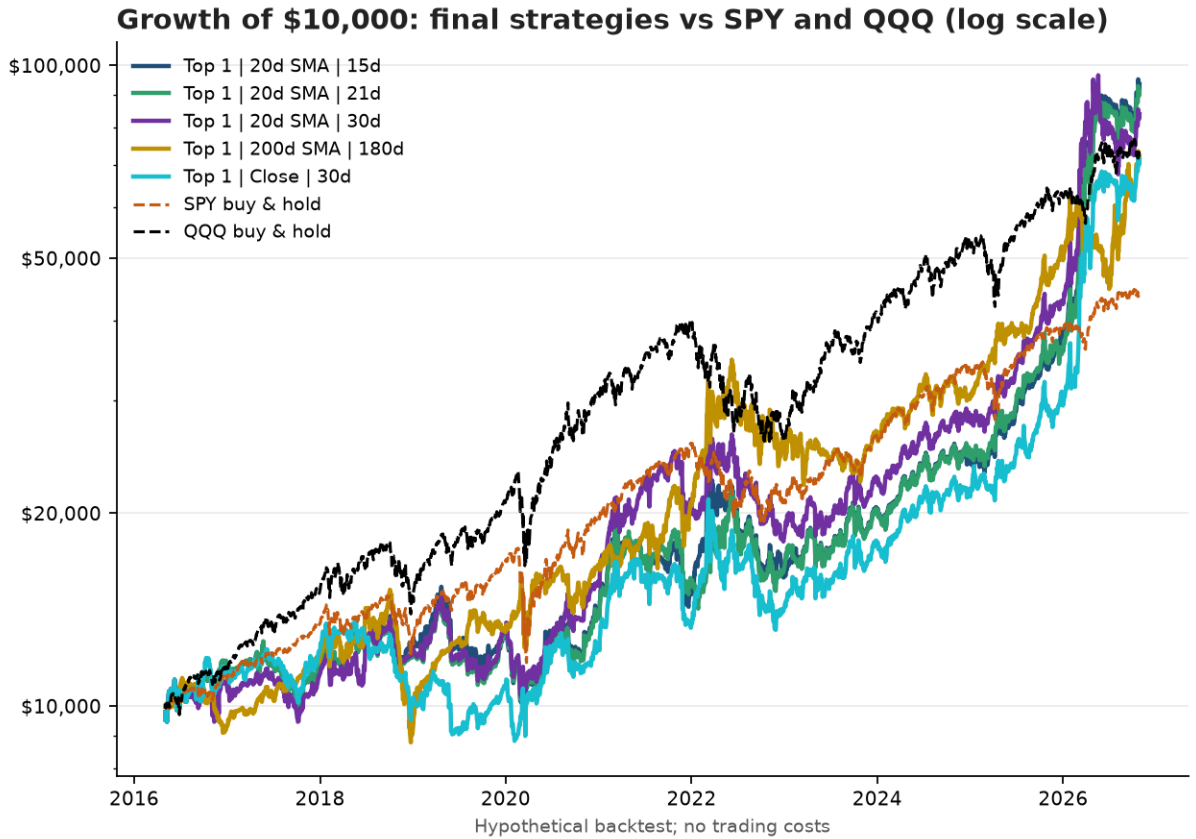
Appendix

Section	Contents	Pages
A. Strategy and Research	Executive summary, strategy, round 1, analysis table 1, PM selection and proposals, decision log, round 2, analysis table 2	14-25
B. Risk and Portfolio Fit	Risk slides and the diversification section	26-29
C. Process and Governance	The strategy meeting, and the full harness, hooks, guardrails and evals pages	30-32
D. Investment Team	Full profiles of the fictional investment team	33-34
E. Full Disclosures	The full presentation's important disclosures	35

The appendix holds every page of the full presentation not used in the main section, unchanged.

Executive summary: monthly momentum rotation across 7 ETFs

- 89 strategies tested over 2016-05-02 to 2026-10-28: 84 combinations of signal, lookback and portfolio rule in round 1, plus 5 PM proposals in round 2, ranked on total return vs SPY buy & hold.
- Final line-up is five Top 1 strategies, led by Top 1 | 20d SMA | 15d at 834.9% total return (23.8% CAGR) vs SPY at 342.8% (15.3%) and QQQ at 638.3% (21.0%).
- All five beat SPY's 15.3% CAGR, from 20.6% (Top 1 | Close | 30d) to 23.8% (Top 1 | 20d SMA | 15d); only the three 20d SMA variants beat QQQ's 21.0% CAGR.
- Low link to stocks: **correlation with SPY of 0.16** for Top 1 | 20d SMA | 15d, 0.13 for Top 1 | Close | 30d and 0.36 for Top 1 | 200d SMA | 180d, vs 0.93 for QQQ.
- Trade-off: volatility of 26.7% to 27.3% across the five vs SPY's 17.7%, and max drawdowns from -34.4% to -42.2% vs SPY's -33.7%; results lean heavily on 2025 and 2026 YTD.
- Selected with hindsight from many settings tested on one period, with no trading costs modeled: research candidates, not validated rules.

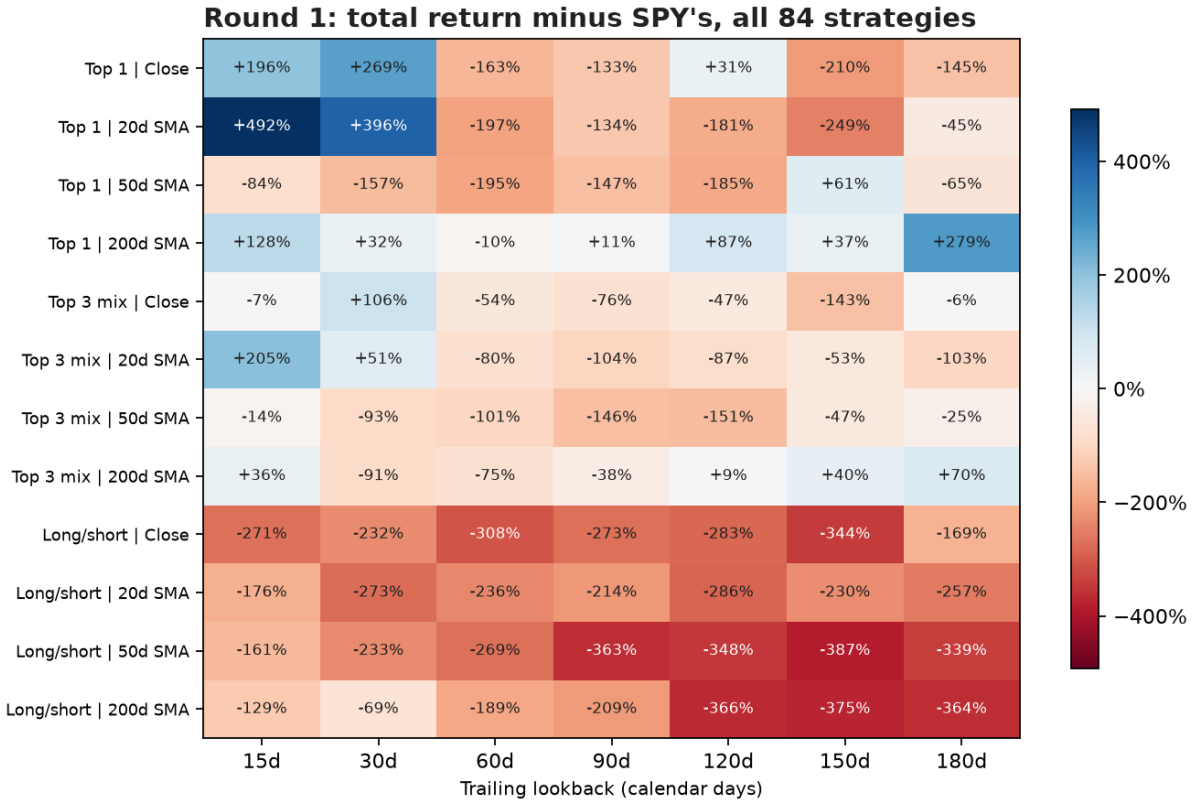


How the strategy works: rank, select, hold for a month

- At each month end, each of 7 ETFs (SPY, QQQ, TLT, IEF, GLD, USO, JNK) is scored by the % change of its signal over a trailing lookback of D calendar days.
- Signals: the raw close, or a 20d, 50d or 200d simple moving average in round 1; round 2 added 10d and 150d SMAs.
- Lookbacks: 15, 30, 60, 90, 120, 150 and 180 calendar days in round 1; round 2 added a 21-day lookback.
- Top 1 holds 100% in the top-ranked ETF; Top 3 mix holds 50%/30%/20% in the top three ETFs.
- Long top 3, short negatives adds an equal-weight short (100% in total) of every other ETF with a negative score.
- Holdings are bought at the month-end close and drift until the next month end; no trading costs, fees or taxes are modeled.

Round 1: what the 84 strategies showed

- Pro: 19 of 84 strategies beat SPY's 342.8%; Top 1 | 20d SMA | 15d led with 834.9% (CAGR 23.8%), ahead of QQQ's 638.3%, with a **correlation with SPY of 0.16**. Only 2 of 84 beat QQQ.
- Pro: Top 3 mix | 20d SMA | 15d had the best risk profile: Sharpe 1.01, 16.8% volatility and a -27.5% max drawdown vs SPY's 0.76, 17.7% and -33.7%.
- Pro: The 200d SMA signal had the highest median total return (332.5%) and 10 strategies beating SPY; the 15d lookback median was 332.2%.
- Con: Top 1 winners carry about 27% volatility; the leader held USO in 36.5% of months with 89 leader changes, so results lean on oil timing.
- Con: Results are sensitive to settings: no 60d strategy beat SPY, and the 50d SMA signal had only 1 strategy beating SPY.
- Con: Long top 3, short negatives failed: median 72.6% and none beat SPY; the worst (50d SMA | 150d) lost -44.3% with a -75.1% max drawdown.



Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

Round 1 by portfolio rule

- Pro: Top 3 mix had the highest median total return of the three rules (292.9%), with 7 strategies beating SPY; its best was Top 3 mix | 20d SMA | 15d.
- Pro: Top 1 had a 287.9% median and the most strategies beating SPY (12), led by Top 1 | 20d SMA | 15d.
- Con: Long top 3, short negatives had a median of only 72.6% and 0 strategies beating SPY.
- Con: 84 combinations were tested on one historical period with no trading costs, so top-ranked results likely include selection bias and may not hold out of sample.

Analysis Table 1 of 2: Round 1, All 84 Strategies

Total return by lookback (days)	15	30	60	90	120	150	180
Top 1 | Close	539.1%	**611.9%**	179.7%	209.7%	374.0%	132.5%	197.5%
Top 1 | 20d SMA	**834.9%**	**739.0%**	145.8%	209.1%	161.4%	93.8%	297.7%
Top 1 | 50d SMA	259.1%	185.9%	148.0%	195.7%	158.1%	403.9%	278.1%
Top 1 | 200d SMA	470.7%	375.0%	332.5%	353.6%	429.4%	380.2%	**622.1%**
Top 3 mix | Close	335.9%	448.7%	288.5%	267.1%	295.6%	200.1%	336.8%
Top 3 mix | 20d SMA	**548.1%**	394.1%	262.9%	238.4%	255.8%	290.3%	239.9%
Top 3 mix | 50d SMA	328.6%	249.7%	241.7%	196.4%	191.6%	295.4%	317.8%
Top 3 mix | 200d SMA	379.0%	251.6%	268.0%	305.1%	351.7%	382.9%	412.7%
Long top 3, short negatives | Close	71.7%	110.6%	35.1%	69.9%	59.7%	-1.1%	173.6%
Long top 3, short negatives | 20d SMA	167.0%	69.4%	107.1%	129.2%	56.9%	112.9%	85.7%
Long top 3, short negatives | 50d SMA	181.4%	110.1%	73.4%	-20.2%	-5.3%	-44.3%	3.7%
Long top 3, short negatives | 200d SMA	213.8%	273.7%	153.5%	134.3%	-23.4%	-32.6%	-21.3%

Max drawdown by lookback (days)	15	30	60	90	120	150	180
Top 1 | Close	-32.6%	**-37.5%**	-40.2%	-37.5%	-32.5%	-50.2%	-47.3%
Top 1 | 20d SMA	**-38.2%**	**-34.4%**	-44.8%	-40.5%	-53.4%	-43.8%	-50.5%
Top 1 | 50d SMA	-41.9%	-50.3%	-37.5%	-45.9%	-43.1%	-40.0%	-48.8%
Top 1 | 200d SMA	-44.0%	-43.3%	-51.1%	-47.0%	-42.2%	-42.2%	**-42.2%**
Top 3 mix | Close	-31.2%	-28.9%	-27.7%	-21.7%	-23.5%	-30.5%	-29.5%
Top 3 mix | 20d SMA	**-27.5%**	-26.1%	-26.1%	-26.1%	-36.4%	-33.6%	-38.5%
Top 3 mix | 50d SMA	-26.2%	-26.4%	-28.3%	-36.0%	-35.9%	-30.5%	-31.9%
Top 3 mix | 200d SMA	-29.9%	-30.9%	-32.0%	-32.0%	-32.0%	-32.0%	-32.0%
Long top 3, short negatives | Close	-46.3%	-44.4%	-46.2%	-36.0%	-44.2%	-53.7%	-42.0%
Long top 3, short negatives | 20d SMA	-38.7%	-38.6%	-37.4%	-39.6%	-47.5%	-44.8%	-41.9%
Long top 3, short negatives | 50d SMA	-35.8%	-37.5%	-40.8%	-62.4%	-55.2%	-75.1%	-71.2%
Long top 3, short negatives | 200d SMA	-42.7%	-39.1%	-45.2%	-47.9%	-73.0%	-75.9%	-78.8%

Correlation with SPY by lookback (days)	15	30	60	90	120	150	180
Top 1 | Close	0.15	**0.13**	0.17	0.16	0.13	0.32	0.31
Top 1 | 20d SMA	**0.16**	**0.16**	0.35	0.33	0.31	0.30	0.29
Top 1 | 50d SMA	0.23	0.36	0.35	0.34	0.30	0.35	0.34
Top 1 | 200d SMA	0.22	0.20	0.32	0.34	0.34	0.34	**0.36**
Top 3 mix | Close	0.31	0.29	0.32	0.31	0.34	0.44	0.45
Top 3 mix | 20d SMA	**0.31**	0.37	0.47	0.45	0.51	0.51	0.53
Top 3 mix | 50d SMA	0.41	0.46	0.51	0.52	0.56	0.58	0.58
Top 3 mix | 200d SMA	0.46	0.47	0.52	0.53	0.54	0.54	0.55
Long top 3, short negatives | Close	-0.07	-0.14	-0.11	-0.13	-0.11	-0.02	-0.02
Long top 3, short negatives | 20d SMA	-0.11	-0.07	-0.02	-0.03	0.02	0.03	0.08
Long top 3, short negatives | 50d SMA	-0.06	-0.01	0.03	0.18	0.22	0.24	0.15
Long top 3, short negatives | 200d SMA	0.01	-0.01	0.03	0.07	0.11	0.11	0.13

Common window 2016-05-02 to 2026-10-28. Benchmarks: SPY total return 342.8%, max drawdown -33.7%; QQQ 638.3%, -35.1%, correlation with SPY 0.93. Correlation is of daily returns. Bold: the round-1 top 5 by total return. Signal = the close or an N-day moving average; lookback = calendar days of % change ranked at each month end. No trading costs.

PM selection: the round 1 top 5 by total return

- The PM ranked all 84 strategies on total return vs SPY and kept the five highest; the long/short family was dropped.
- Top 1 | 20d SMA | 15d (834.9%) and Top 1 | 20d SMA | 30d (739.0%) ranked first and second, suggesting the short-lookback 20d SMA result is not a one-off.
- Top 1 | 200d SMA | 180d (622.1%) ranked third: a slow trend signal that switched leaders only 16 times, but with a -42.2% drawdown.
- Top 1 | Close | 30d (611.9%) ranked fourth, with the lowest corr_spy in the top 5 at 0.13.
- Top 3 mix | 20d SMA | 15d (548.1%) ranked fifth, with the best risk-adjusted profile of the leaders (Sharpe 1.01).

Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

PM proposals: five neighbours of the round 1 leaders

- Top 1 | 20d SMA | 21d: sits between the two strongest 20d SMA lookbacks (15d and 30d), testing whether that region holds up at nearby settings.
- Top 1 | 10d SMA | 15d: checks whether the edge comes from short-term smoothing in general or from the 20d SMA specifically.
- Top 3 mix | 20d SMA | 21d: a diversified version of the short-lookback 20d SMA signal, aiming to keep a Sharpe near the 1.01 of Top 3 mix | 20d SMA | 15d.
- Top 1 | 150d SMA | 180d: tests whether the slow-trend result of Top 1 | 200d SMA | 180d holds up with slightly faster smoothing.
- Top 1 | Close | 21d: the simplest signal, near the strong Close | 30d and Close | 15d results, with no smoothing parameter to overfit.

Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

Decision Log, Round 1: Research Findings and the PM's Top 5

Research Agent to Portfolio Manager

Portfolio Manager Agent: the short-lookback 20d SMA signals led on return, with Top 1 | 20d SMA | 15d at 834.9% but carrying 27.1% volatility and a -38.2% drawdown, while Top 3 mix | 20d SMA | 15d gave the best Sharpe (1.01) with a -27.5% drawdown. The long/short family underperformed across the board and looks worth dropping. For round 2, consider probing neighbouring parameters around the 15-30d lookbacks and the 200d SMA / 180d region, favouring Top 3 mix for robustness, and keep in mind the dependence on USO and the single-period, many-trials caveat.

- Pro: Top 1 | 20d SMA | 15d led with 834.9% total return (CAGR 23.8%) vs SPY 342.8% and QQQ 638.3% over 2016-05-02 to 2026-10-28, with low SPY **correlation (0.16)**; only 2 of 84 strategies beat QQQ and 19 beat SPY.
- Pro: Top 3 mix | 20d SMA | 15d had the best risk profile among leaders: 548.1% return, Sharpe 1.01, max drawdown -27.5% and volatility 16.8%, versus SPY's Sharpe 0.76, drawdown -33.7% and volatility 17.7%.
- Con: Top 1 winners carry higher risk: volatility around 27% (27.1% for the leader) and deeper drawdowns (-38.2% for the leader, -42.2% for Top 1 | 200d SMA | 180d) than SPY's -33.7%; the leader also held USO in 36.5% of months with 89 leader changes, so results lean heavily on oil timing.
- Con: Results are sensitive to lookback: 15d (median 332.2%) and 180d (259.0%) did well, but no 60d strategy beat SPY; the 200d SMA signal had the highest median (332.5%) while 50d SMA had only 1 strategy beating SPY.
- Con: Long top 3, short negatives failed: median 72.6%, none beat SPY, and the worst (50d SMA | 150d) lost -44.3% with a -75.1% max drawdown, averaging 2.0 ETFs shorted.
- Con: 84 combinations were tested on one historical period with no trading costs, so top-ranked results likely include selection bias and may not hold out of sample.

PM: top 5 selected (by total return)

Top 1 | 20d SMA | 15d (834.9%): Top 1 | 20d SMA | 15d ranks first with 834.9% total return, 492.1% ahead of SPY's 342.8%. **Pro:** Low SPY **correlation (0.16)** and a 23.8% CAGR, also ahead of QQQ's 638.3%. **Con:** 27.1% volatility and a -38.2% max drawdown, with USO leading in 36.5% of months over 89 leader changes.

Top 1 | 20d SMA | 30d (739.0%): Top 1 | 20d SMA | 30d ranks second with 739.0% total return, which suggests the short-lookback 20d SMA result is not a one-off. **Pro:** Its -34.4% max drawdown is the shallowest of the Top 1 leaders, with a Sharpe of 0.80 and corr_spy of 0.16. **Con:** Volatility of 27.1% and 91 leader changes, with USO leading in 36.5% of months.

Top 1 | 200d SMA | 180d (622.1%): Top 1 | 200d SMA | 180d ranks third with 622.1%, using a slow trend signal that switched leaders only 16 times. **Pro:** Only 16 leader changes, so it trades much less than the 20d SMA leaders. **Con:** Deepest drawdown in the top 5 at -42.2%, and the highest corr_spy among them at 0.36.

Top 1 | Close | 30d (611.9%): Top 1 | Close | 30d ranks fourth with 611.9%, showing that a raw-close signal also works at short lookbacks. **Pro:** Lowest corr_spy in the top 5 at 0.13, and 269.1% ahead of SPY. **Con:** USO led in 37.3% of months, and its -37.5% drawdown is deeper than SPY's -33.7%.

Top 3 mix | 20d SMA | 15d (548.1%): Top 3 mix | 20d SMA | 15d ranks fifth with 548.1% and has the best risk-adjusted profile of the leaders. **Pro:** Sharpe of 1.01 against SPY's 0.76, with a -27.5% drawdown and 16.8% volatility, both below SPY's. **Con:** Its 548.1% return trails QQQ's 638.3%, and it still relies on the same USO-heavy ranking (USO led 36.5% of months).

Decision Log, Round 1: The PM's Proposals to the Research Agent

Portfolio Manager to Research Agent

> Research Agent: please run these five neighbours of the round 1 leaders. They focus on the 15-30d lookbacks with 20d/10d SMA or the close, plus one slower 150d SMA | 180d variant. Long/short is dropped. Please report USO leadership share and leader changes for each.

Each proposal adjusts a parameter of a strategy that worked; the Research Agent ran all of them on the same dates as round 1.

PM: 5 new parameter sets

Top 1 | 20d SMA | 21d: A 21d lookback sits between the two strongest 20d SMA lookbacks (15d and 30d), so it tests whether that region holds up across nearby settings. **Pro:** Tests robustness inside the best-performing region rather than at an extreme. **Con:** Concentrated in one ETF, so volatility will likely stay as high as the Top 1 leaders'.

Top 1 | 10d SMA | 15d: A shorter 10d SMA with the 15d lookback may react faster than the 20d SMA leader while still smoothing noise. **Pro:** Checks whether the edge comes from short-term smoothing in general or from 20d specifically. **Con:** Faster signals may lead to more leader changes and more USO dependence.

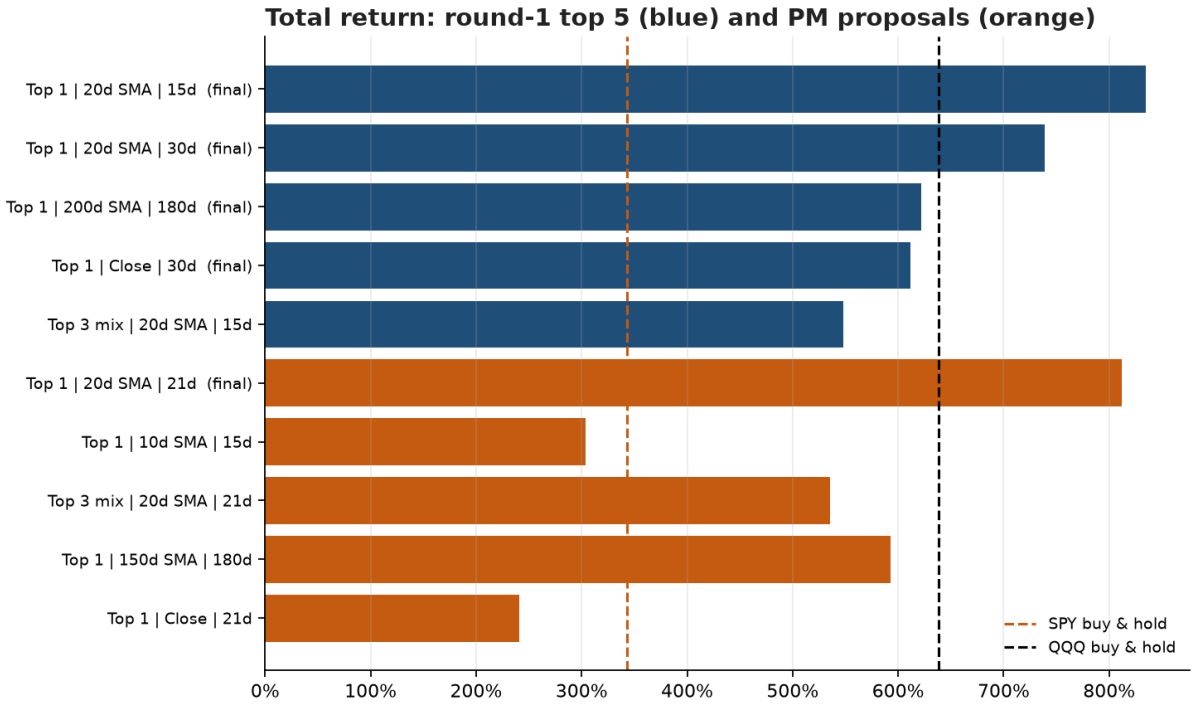
Top 3 mix | 20d SMA | 21d: Diversified version of the 20d SMA short-lookback signal at a middle lookback; it may keep a Sharpe near Top 3 mix | 20d SMA | 15d's 1.01. **Pro:** Spreading across three ETFs has historically cut volatility to about 17%. **Con:** Probably lower total return than the Top 1 variants, so it may not enter the top 5.

Top 1 | 150d SMA | 180d: A 150d SMA with the 180d lookback tests whether the slow-trend result of Top 1 | 200d SMA | 180d holds up with a slightly faster smoothing window. **Pro:** Likely few leader changes, so the result is less exposed to the missing trading-cost assumption. **Con:** Top 1 | 200d SMA | 180d already had a -42.2% drawdown; a slow signal may react late in reversals.

Top 1 | Close | 21d: A raw-close signal over 21d, near the strong Close | 30d and Close | 15d results, tests the close-based short-momentum region. **Pro:** Simplest signal, with no smoothing parameter to overfit. **Con:** Raw-close signals are noisier and may switch to USO even more often.

Round 2: one replacement in the top 5

- Two proposals beat the weakest round 1 member, Top 3 mix | 20d SMA | 15d (548.1%): Top 1 | 20d SMA | 21d (811.9%) and Top 1 | 150d SMA | 180d (592.5%).
- Top 1 | 20d SMA | 21d replaced Top 3 mix | 20d SMA | 15d and entered at #2, with a Sharpe of 0.82 and a -34.4% max drawdown.
- Top 1 | 150d SMA | 180d (592.5%) fell short of the new #5, Top 1 | Close | 30d (611.9%), so only one replacement was made.
- Top 3 mix | 20d SMA | 21d returned 535.8%, below the cut, but kept a Sharpe of 1.00 and the shallowest drawdown tested (-23.2%).
- Sensitivity: Top 1 | 10d SMA | 15d returned 303.6% and Top 1 | Close | 21d 240.7%, both trailing SPY's 342.8%.



Decision Log, Round 2: Proposal Results and the PM's Final Decisions

Research Agent to Portfolio Manager

Portfolio Manager Agent: of your five proposals, Top 1 | 20d SMA | 21d (811.9%) and Top 1 | 150d SMA | 180d (592.5%) beat the round-1 #5 on total return. Top 3 mix | 20d SMA | 21d fell just short on return but had a Sharpe of 1.00 and a -23.2% drawdown. The 10d SMA | 15d and Close | 21d variants trailed SPY, which shows strong sensitivity to the signal and lookback choices. Given the high USO leadership share (37.3%-38.9% of months for the short-lookback variants) and the many settings tested on one period, consider out-of-sample or sub-period checks and trading-cost assumptions before favoring any single setting.

- Pro: Top 1 | 20d SMA | 21d returned 811.9% (excess vs SPY 469.1%, Sharpe 0.82, max drawdown -34.4%), landing between the round-1 #1 (834.9%) and #2 (739.0%), so the 20d SMA 15-30d region held up at a nearby lookback.
- Pro: Top 3 mix | 20d SMA | 21d kept a Sharpe of 1.00 with 16.8% volatility and the shallowest drawdown tested (-23.2%), although its 535.8% total return is below the round-1 #5 (548.1%).
- Pro: Top 1 | 150d SMA | 180d returned 592.5% (Sharpe 0.76) with only 16 leader changes, so it is less exposed to the missing trading costs. It also beats the round-1 #5 on total return.
- Con: Results are very sensitive to the smoothing choice. Top 1 | 10d SMA | 15d returned 303.6% (excess vs SPY -39.2%, Sharpe 0.56), against 834.9% for the 20d SMA at the same lookback.
- Con: Top 1 | Close | 21d returned only 240.7% (excess vs SPY -102.1%, Sharpe 0.49, drawdown -40.0%), against 611.9% for Close | 30d, so the raw-close result looks fragile to the lookback.
- Con: USO dependence stays high: USO led 38.1% of months for both 20d SMA | 21d variants, 38.9% for 10d SMA | 15d, 37.3% for Close | 21d and 28.6% for 150d SMA | 180d. The short-lookback variants switched leaders often (93 changes for 20d SMA | 21d, 101 for Close | 21d).
- Con: Top 1 | 150d SMA | 180d matched the -42.2% max drawdown of Top 1 | 200d SMA | 180d. Many combinations have now been tested on one historical period, so data-mining risk is growing.

PM: final decisions

The final top 5 is four Top 1 strategies on short-lookback 20d SMA or raw-close signals plus one slow 200d SMA | 180d strategy. All five beat SPY buy & hold's 342.8% in this window, but each has higher volatility than SPY, and their max drawdowns run from -34.4% to -42.2%. The settings were picked from many combinations tested on one historical period with no trading costs, and USO leads often, so treat these as research candidates. They are not validated rules. The next step should be sub-period and out-of-sample checks plus cost assumptions, with the lower-volatility Top 3 mix variants kept as reference points.

[KEPT] Top 1 | 20d SMA | 15d (834.9%): Kept at #1 because it has the highest total return in the final top 5: 834.9%, which is 492.1% ahead of SPY buy & hold (342.8%). **Pro:** Highest total return and the highest Sharpe among the final five (0.84), with a corr_spy of 0.16. **Con:** It holds a single ETF with 27.1% volatility and a -38.2% max drawdown, deeper than SPY's -33.7%. USO led 36.5% of months, and 89 leader changes were made with no trading costs modeled.

[ADDED] Top 1 | 20d SMA | 21d (811.9%): Added at #2 under the replacement rule: its 811.9% total return beat the weakest round-1 member, Top 3 mix | 20d SMA | 15d (548.1%). It also shows the 20d SMA short-lookback region holds up at a nearby lookback. **Pro:** Excess return vs SPY of 469.1% with a Sharpe of 0.82, and a -34.4% max drawdown that is shallower than Top 1 | 20d SMA | 15d's -38.2%. **Con:** Highest volatility in the final five (27.3%). USO led 38.1% of months, and its 93 leader changes make it the most exposed to the trading costs left out of the model.

[KEPT] Top 1 | 20d SMA | 30d (739.0%): Kept at #3 with 739.0% total return (excess vs SPY 396.2%), giving a third data point in the same 20d SMA region. **Pro:** Its -34.4% max drawdown ties Top 1 | 20d SMA | 21d for the shallowest in the final five, and its Sharpe is 0.80. **Con:** Volatility is 27.1%, its drawdown is still deeper than SPY's -33.7%, USO led 36.5% of months, and it made 91 leader changes with no costs modeled.

[KEPT] Top 1 | 200d SMA | 180d (622.1%): Kept at #4 with 622.1% total return (excess vs SPY 279.3%). It is the slow-signal alternative to the 20d SMA cluster. **Pro:** Only 16 leader changes, so it is the least exposed to the missing trading costs. QQQ led most often (34.9%), not USO (26.2%). **Con:** Deepest max drawdown in the final five (-42.2%), the highest corr_spy (0.36) and a Sharpe of 0.75, below SPY's 0.76.

[KEPT] Top 1 | Close | 30d (611.9%): Kept at #5 with 611.9% total return (excess vs SPY 269.1%). Top 1 | 150d SMA | 180d (592.5%) did not beat it, so it stays. **Pro:** Lowest corr_spy in the final five (0.13), which gives the most diversification against SPY. **Con:** Fragile to the lookback: Top 1 | Close | 21d returned only 240.7% (excess vs SPY -102.1%). It also has a -37.5% max drawdown, 26.7% volatility, USO leading 37.3% of months and 90 leader changes.

[REMOVED] Top 3 mix | 20d SMA | 15d (548.1%): Removed under the replacement rule: its 548.1% total return was the weakest of the round-1 top 5, and Top 1 | 20d SMA | 21d beat it with 811.9%. **Pro:** Best risk profile of the group: Sharpe of 1.01, 16.8% volatility (below SPY's 17.7%) and a -27.5% max drawdown. **Con:** Its 548.1% total return trails QQQ buy & hold (638.3%) and every member of the final five, and the objective ranks on total return.

Analysis Table 2 of 2: Portfolio Manager Rounds and the Final Line-up

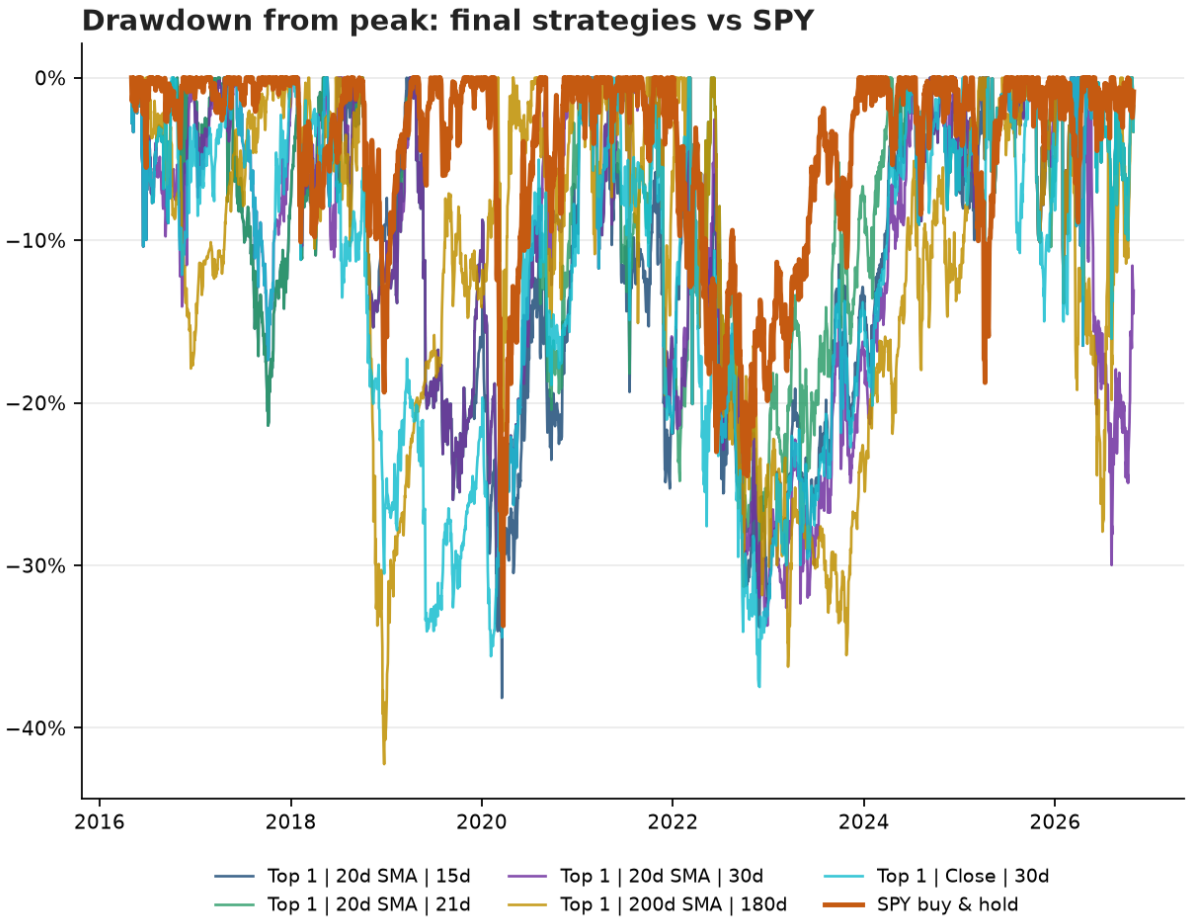
Strategy	Round	Status	Total return	vs SPY	Ann. return	Max DD	Volatility	Sharpe	Corr. SPY
Top 1 | 20d SMA | 15d	**1: top 5**	**Final**	**834.9%**	**492.1%**	**23.8%**	**-38.2%**	**27.1%**	**0.84**	**0.16**
Top 1 | 20d SMA | 30d	**1: top 5**	**Final**	**739.0%**	**396.2%**	**22.5%**	**-34.4%**	**27.1%**	**0.80**	**0.16**
Top 1 | 200d SMA | 180d	**1: top 5**	**Final**	**622.1%**	**279.3%**	**20.8%**	**-42.2%**	**26.7%**	**0.75**	**0.36**
Top 1 | Close | 30d	**1: top 5**	**Final**	**611.9%**	**269.1%**	**20.6%**	**-37.5%**	**26.7%**	**0.75**	**0.13**
Top 3 mix | 20d SMA | 15d	1: top 5	Replaced	548.1%	205.2%	19.5%	-27.5%	16.8%	1.01	0.31
Top 1 | 20d SMA | 21d	**2: proposal**	**Final**	**811.9%**	**469.1%**	**23.5%**	**-34.4%**	**27.3%**	**0.82**	**0.16**
Top 1 | 10d SMA | 15d	2: proposal	Not selected	303.6%	-39.2%	14.3%	-35.7%	25.4%	0.56	0.18
Top 3 mix | 20d SMA | 21d	2: proposal	Not selected	535.8%	193.0%	19.3%	-23.2%	16.8%	1.00	0.30
Top 1 | 150d SMA | 180d	2: proposal	Not selected	592.5%	249.7%	20.3%	-42.2%	25.4%	0.76	0.20
Top 1 | Close | 21d	2: proposal	Not selected	240.7%	-102.1%	12.4%	-40.0%	26.0%	0.49	0.16
SPY buy & hold	benchmark		342.8%	0.0%	15.3%	-33.7%	17.7%	0.76	1.00
QQQ buy & hold	benchmark		638.3%	295.5%	21.0%	-35.1%	22.2%	0.86	0.93

Final line-up by period	1Y total	3Y ann.	5Y ann.	SI ann.	Corr. SPY	Corr. TLT	80/20 blend ann.	Blend max DD
Top 1 | 20d SMA | 15d	177.4%	68.7%	39.7%	23.8%	0.16	0.03	17.8%	-28.4%
Top 1 | 20d SMA | 21d	164.2%	67.6%	37.8%	23.5%	0.16	0.02	17.7%	-28.4%
Top 1 | 20d SMA | 30d	110.9%	56.8%	28.0%	22.5%	0.16	0.03	17.5%	-28.4%
Top 1 | 200d SMA | 180d	58.2%	47.8%	27.4%	20.8%	0.36	-0.04	17.0%	-29.7%
Top 1 | Close | 30d	145.6%	60.3%	34.4%	20.6%	0.13	0.06	17.2%	-26.1%
SPY buy & hold	13.6%	25.1%	12.6%	15.3%	1.00		15.3%	-33.7%
QQQ buy & hold	15.2%	29.0%	14.3%	21.0%	0.93	n/a	n/a	n/a

Common window 2016-05-02 to 2026-10-28; periods end on the last completed session. Bold rows: the final line-up. "vs SPY" = total return minus SPY's. 1Y is a total return; longer periods are annualized; SI = the whole window. Blend = 80% SPY + 20% strategy, rebalanced daily; correlation and blend are measured for the final strategies and SPY. Sharpe uses the 13-week T-bill. No trading costs.

Risk: higher volatility and deeper drawdowns than SPY

- Volatility for the final five ran from 26.7% to 27.3% vs 17.7% for SPY and 22.2% for QQQ.
- Max drawdowns ran from -34.4% (Top 1 | 20d SMA | 21d and Top 1 | 20d SMA | 30d) to -42.2% (Top 1 | 200d SMA | 180d), all deeper than SPY's -33.7%.
- Long stretches of lag: in 2017 and 2019 Top 1 | 20d SMA | 15d returned -0.1% and 1.9% vs SPY's 21.7% and 31.2%; Top 1 | Close | 30d lost 16.8% in 2018 vs -4.6% for SPY.
- Recent dependence: 2026 YTD returns were 155.0% for Top 1 | 20d SMA | 15d vs 14.1% for SPY, so the full-period edge is not a steady annual rate.
- No trading costs modeled: Top 1 | 20d SMA | 21d made 93 leader changes vs 16 for Top 1 | 200d SMA | 180d; USO led 36.5% of months for Top 1 | 20d SMA | 15d.



Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

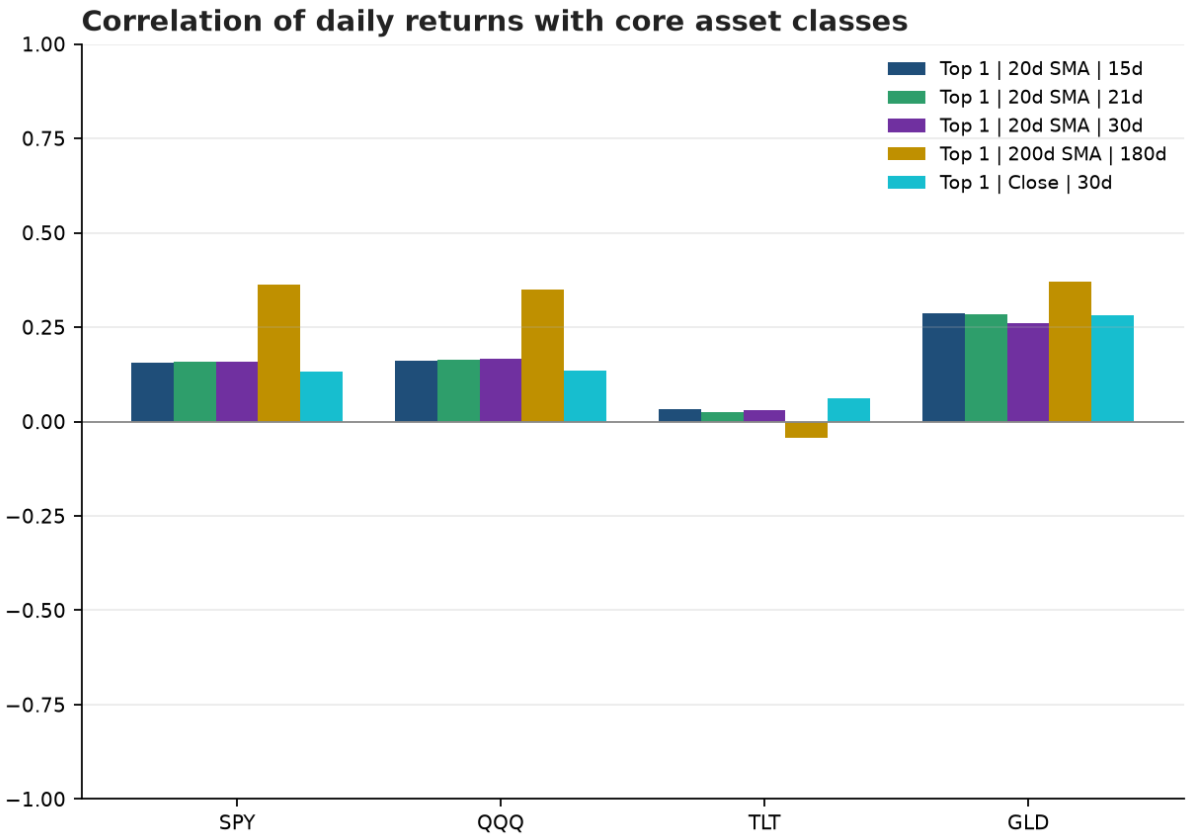
Risk: hindsight selection and the long/short failure

- The final five were selected with hindsight from 89 strategies tested on a single period (2016-05-02 to 2026-10-28); results may not hold out of sample.
- Nearby settings can collapse: Top 1 | Close | 21d returned 240.7% (-102.1% vs SPY) against 611.9% for Top 1 | Close | 30d.
- Long top 3, short negatives: median total return of 72.6% and none of its strategies beat SPY's 342.8%.
- Its worst version, Long top 3, short negatives | 50d SMA | 150d, lost -44.3% with a -75.1% max drawdown; the 200d SMA | 180d version fell as far as -78.8%.
- Next steps planned: sub-period and out-of-sample checks plus trading-cost assumptions.

Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

Diversification: a return stream largely independent of SPY

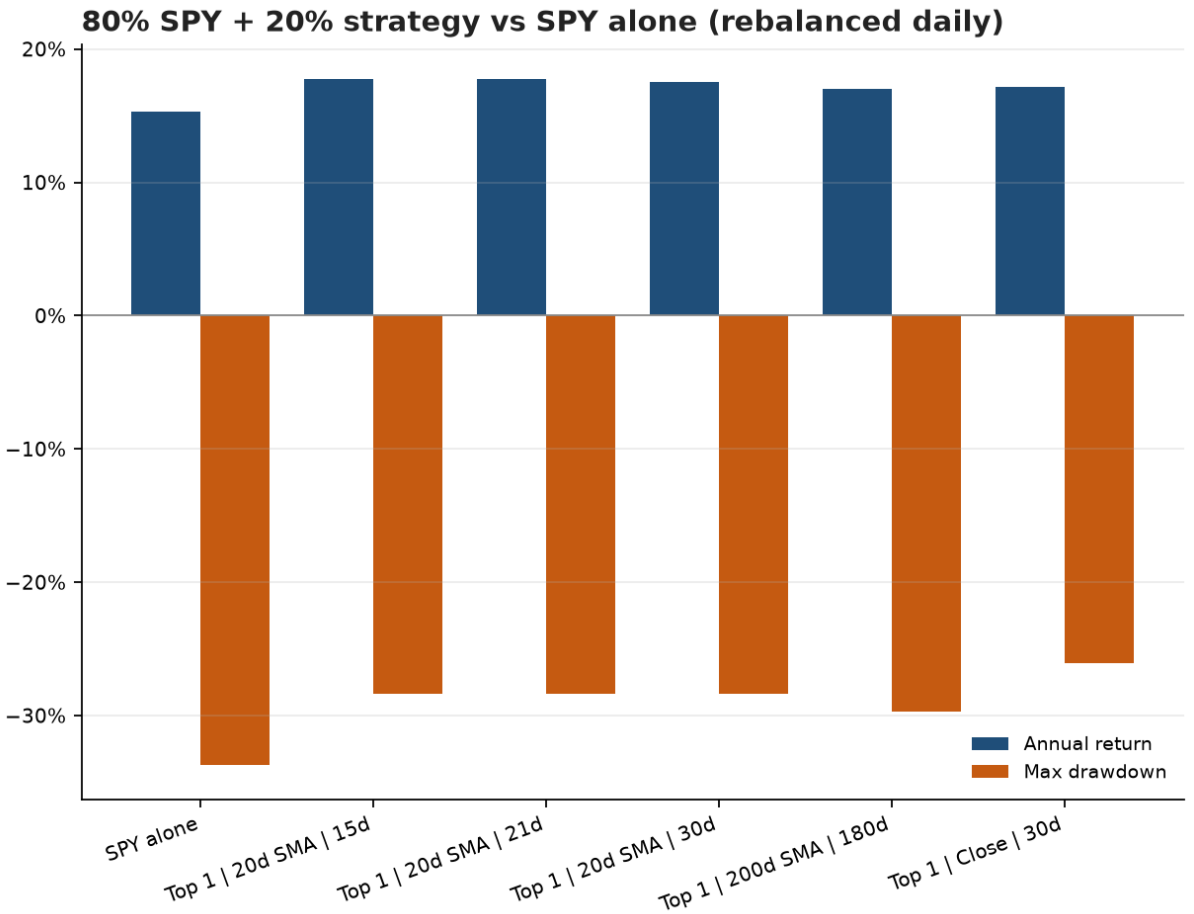
- The core of the case: Top 1 | 20d SMA | 15d has a **correlation with SPY of 0.16** over 2016-05-02 to 2026-10-28, while QQQ, another equity fund, sits at 0.93.
- Across the line-up: **correlation with SPY of 0.13** for Top 1 | Close | 30d, 0.16 for the three 20d SMA variants and 0.36 for Top 1 | 200d SMA | 180d.
- It rotates monthly across stocks, Treasuries, gold, oil and high-yield credit; **correlations with TLT were near zero (-0.04 to 0.06)** across the final five.
- 2022 bear market: Top 1 | 200d SMA | 180d gained 29.0% and Top 1 | 20d SMA | 15d gained 6.4% while SPY fell 18.2% and QQQ 32.6%.
- Investors could view it as a satellite next to a core equity holding, adding a different return stream with a **correlation with SPY of 0.16** (Top 1 | 20d SMA | 15d).
- Balance: results were selected with hindsight on one period, and Top 1 | 20d SMA | 15d lagged SPY in 2017, 2019 and 2024.



Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

Portfolio fit: 80% SPY + 20% strategy

- 80% SPY + 20% Top 1 | 20d SMA | 15d: 17.8% CAGR, 15.9% volatility, 0.96 Sharpe and -28.4% max drawdown vs SPY alone at 15.3%, 17.7%, 0.76 and -33.7%.
- That blend benefits from the strategy's **correlation with SPY of 0.16**: its up capture was 90.0% vs a down capture of 85.8%.
- Every 80/20 blend of the final five had a higher Sharpe than SPY alone, from 0.88 to 0.96 vs 0.76.
- Shallowest blend drawdown: 80% SPY + 20% Top 1 | Close | 30d at -26.1%, drawing on a **correlation with SPY of 0.13**.
- Balance: blends use strategies selected with hindsight, with no trading costs modeled, and Top 1 | 20d SMA | 15d lagged SPY in 2017, 2019 and 2024.



Hypothetical backtested performance, selected with hindsight from many tested combinations. Not actual trading. No trading costs, fees or taxes are included. Past performance does not guarantee future results. See Important Disclosures.

Strategy Meeting: Portfolio Manager, Research and Analytics Agents (simulated conversation)

Simulated conversation, written by the Portfolio Manager Agent from this run's verified results. The speakers are AI agents, not people; every number was checked against the backtest.

Portfolio Manager Agent: Today we compare monthly momentum rotations across SPY, QQQ, TLT, IEF, GLD, USO and JNK against SPY buy & hold, over 2016-05-02 to 2026-10-28. The objective is the highest total return relative to SPY, and we look at drawdown, volatility and Sharpe alongside it.

Analytics Agent: The benchmark is SPY buy & hold at 342.8%, with a -33.7% max drawdown, 17.7% volatility and a Sharpe of 0.76. QQQ buy & hold returned 638.3%, so it is a tougher reference point.

Research Agent: In round 2, Top 1 | 20d SMA | 21d returned 811.9% and replaced Top 3 mix | 20d SMA | 15d, which had 548.1%. Top 1 | 150d SMA | 180d reached 592.5%, but that fell short of the new #5, Top 1 | Close | 30d at 611.9%.

Portfolio Manager Agent: Our final #1 is Top 1 | 20d SMA | 15d at 834.9%, or 492.1% ahead of SPY. Three of the top five use the 20d SMA with lookbacks of 15, 21 and 30 days, so the result is not resting on a single lookback.

Analytics Agent: The risk is real, though. All five Top 1 strategies run at 26.7% to 27.3% volatility, against SPY's 17.7%. Their max drawdowns run from -34.4% for Top 1 | 20d SMA | 21d and Top 1 | 20d SMA | 30d to -42.2% for Top 1 | 200d SMA | 180d, and every one is deeper than SPY's -33.7%.

Research Agent: Concentration is the other issue. USO led 38.1% of months for Top 1 | 20d SMA | 21d and 36.5% for Top 1 | 20d SMA | 15d. Nearby settings can collapse, too: Top 1 | Close | 21d returned 240.7%, which is -102.1% versus SPY.

Analytics Agent: Turnover matters as well, because no trading costs are modeled. Top 1 | 20d SMA | 21d made 93 leader changes, while Top 1 | 200d SMA | 180d made only 16.

Portfolio Manager Agent: That is why Top 1 | 200d SMA | 180d stays as the slow-signal alternative at 622.1%, even with its -42.2% drawdown. And Top 1 | Close | 30d gives the lowest corr_spy at 0.13.

Research Agent: The removed Top 3 mix | 20d SMA | 15d still had the best risk profile: a Sharpe of 1.01, 16.8% volatility and a -27.5% drawdown. I'd keep it as a reference point.

Portfolio Manager Agent: Agreed. Next we run sub-period and out-of-sample checks and add cost assumptions. We have tested many combinations on one period, so these are candidates and not conclusions.

The Harness: Orchestration, Hooks and Cost Tracking

Harness

• Deterministic Python orchestrates the agents; no LLM decides the order of work.

• Each stage runs one agent, then a validator checks its saved artifact before the next stage starts.

• Agent loop: Claude Messages API with prompt caching, concurrent tool calls, a turn limit per agent and a token budget per run; tool errors go back to the agent to fix.

• Trace: every model call and tool call is logged (trace.jsonl) with tokens, latency and results.

• Cost: tokens and dollars are attributed to each agent and each tool; the manifest records both.

• Artifacts: every stage saves JSON/CSV, so any stage, or just this PDF, can be rebuilt from a finished run without re-running the agents (build-pdf).

Hooks

forbid_delete (pre-tool): Denies any tool whose name implies deletion and any input carrying a delete command (rm, del, Remove-Item, os.remove, shutil.rmtree, .unlink()).

confine_paths (pre-tool): Denies any path-like input that resolves outside the project folder (absolute paths elsewhere, ~, or .. traversal).

count_calls (post-tool): Counts calls and failures per agent and tool for the run manifest.

audit hook (process): A Python audit hook blocks file deletion, writes outside the project and reads outside the project and the Python installation, whatever code asks for it.

guard.py (Claude Code): A PreToolUse hook stops developer sessions from deleting files or reading, writing or searching outside the project.

This run: tokens and cost

Total 386,383 tokens, $1.20, 323 seconds, model claude-opus-5-5.

Agent	Input tokens	Output tokens	Cost
Presentation Agent	113,704	11,260	$0.407
Portfolio Manager Agent	134,123	8,068	$0.346
Research Agent	56,998	4,622	$0.186
Analytics Agent	36,997	3,675	$0.143
Team Agent	5,824	2,712	$0.076
Data Agent	7,207	1,193	$0.037

Cost owner (tool or context)	Input tokens	Output tokens	Cost
add_slide	12,293	9,568	$0.203
(system & tool definitions)	82,370	0	$0.178
get_results	68,507	94	$0.087
list_metrics	49,241	27	$0.079
(agent text)	72	3,826	$0.077
save_research	4,940	3,213	$0.068
submit_team	3,019	2,557	$0.061
get_dialogue	33,210	27	$0.053
save_analytics	4,260	2,395	$0.051
strategy_detail	15,180	1,341	$0.041

Top 10 cost owners. Input tokens are attributed to the content each call re-read (a tool owns its requests and results), output tokens to the blocks written; each share is priced at that call's rates, so all owners together add up to the run total (the manifest lists every owner).
Hook denials this run: 0.

Guardrails and Evals

Guardrails

Numeric faithfulness. Every number an agent writes must match a computed result or a configured parameter at the rounding used. Violations go back to the agent to fix.

Compliance language. Promissory, risk-denying and urgency phrases are blocked (for example 'guaranteed', 'can't lose', 'act now').

Decisions by construction. The top-5 ranking and the replacement rule are computed in code. The Portfolio Manager's choices are accepted only when they match.

Comparable tests. Every strategy is scored on the same dates. Duplicate, out-of-range or too-short-history proposals are rejected.

Fictional personas. Team members are AI-generated, labelled fictional on every card, and never named on agent slides.

Ticker allowlist. Only the seven ETFs and the T-bill rate can be fetched.

Filesystem confinement. All reads and writes go through one confinement function; nothing is ever deleted.

Stage validators. Each stage's output is checked before the next agent starts; every slide is re-checked before the PDF is built.

Evals: 15 scorers on every finished run

run_health: The run finished, every agent replied, no tool crashed, cost under the cap.

data_integrity: All seven ETFs and the T-bill rate present, positive, recent and in date order.

research_coverage: Round 1 ran all 84 strategies; round 2 ran every proposal.

research_reproducible: Recomputing every strategy from the saved prices gives identical returns and scores.

notebook_crosscheck: An independent implementation copied from the research notebook matches every strategy.

pm_protocol: Top 5 by total return, 5 valid new proposals, replacement rule applied correctly.

dialogue_complete: Research and PM messages for both rounds, decisions with pros and cons, the meeting, and the constructive and harsh feedback scripts.

narrative_faithfulness: Every number in agent-written text is found in the computed results.

compliance_language: No promissory or urgency language; no persona names on agent slides.

cost_accounting: Per-tool and per-agent costs add up to the run's total cost.

hook_enforcement: Delete and path-confinement hooks were installed on every agent and block red-team inputs.

pdf_structure: Title, agent-process first page, flowchart, governance pages, meeting, constructive and harsh feedback pages, decision log, at most 2 analysis-table pages, diversification section, team and disclosures.

pdf_disclosures: Performance footer on every performance page, cost and hypothetical disclosures present.

team_quality: Six plausible fictional personas, each labelled.

pitchbook_structure: Pitchbook main pages in order (summary, performance, team, investment process, feedback, costs per tool, governance, compliance, disclosures), costs matching the manifest, and every unused presentation page in its appendix.

Other eval layers

• Unit tests for the engine, agents (with a scripted Claude client), hooks, sandbox, PDF and evals.

• Red-team suite: invented numbers, promissory language, rule-breaking PM inputs, delete commands and out-of-project paths must all be blocked; benign look-alikes must pass.

• Live trials: the full pipeline is run several times and pass rates per scorer are reported.

Meet the Digital Investment Team (AI-generated, fictional)

Margaret Ellison-Hart

Lead Portfolio Manager

Focus: Strategy oversight, allocation decisions and investment committee

Certifications: CFA, CAIA | **Experience:** 26 yrs

Education

BA, Economics, University of Michigan (2000)
MBA, Finance, University of Chicago Booth School of Business (2006)

Experience

Senior Portfolio Manager, Tactical Allocation, Harborview Capital Partners (2013-2024)
Associate Portfolio Manager, Northgate Asset Strategies (2006-2013)
Research Associate, Clearwater Fiscal Advisors (2000-2004)

Margaret Ellison-Hart has spent more than two decades running tactical and multi-asset portfolios, beginning as a research associate before leading allocation teams at Harborview Capital Partners. On this team she owns the overall strategy, chairs the monthly investment committee, and makes the final call on rule changes, ensuring every adjustment is documented and justified before implementation. Her distinctive strength is discipline: she is known for protecting systematic processes from discretionary second-guessing while still asking hard questions about whether a signal's logic continues to hold across changing market regimes.

AI-generated fictional persona

Daniel Okafor

Senior Quantitative Analyst

Focus: Quantitative research and signal design

Certifications: CQF | **Experience:** 12 yrs

Education

BSc, Mathematics, University of Toronto (2008)
PhD, Statistics, Carnegie Mellon University (2014)

Experience

Quantitative Researcher, Meridian Ridge Quantitative (2014-2021)
Senior Quant Analyst, Silverpine Systematic Partners (2021-2025)

Daniel Okafor is a statistician who moved from academic research into systematic investing, building momentum and trend models at Meridian Ridge Quantitative and Silverpine Systematic Partners. On this team he owns signal design, including the choice between raw price change and moving-average change, lookback windows, and ranking rules. His distinctive strength is rigor around overfitting: he insists on out-of-sample testing, parameter stability checks and plain-language explanations of why a signal should work before it earns a place in the model.

AI-generated fictional persona

Priya Raman

Risk Management Analyst

Focus: Risk management, drawdown monitoring and position concentration

Certifications: FRM, PRM | **Experience:** 13 yrs

Education

BTech, Industrial Engineering, Indian Institute of Technology Bombay (2011)
MS, Financial Engineering, Columbia University (2013)

Experience

Risk Analyst, Bastion Lane Risk Advisory (2013-2018)
Senior Risk Analyst, Cedarstone Multi-Asset Group (2018-2025)

Priya Raman trained as an engineer before specializing in financial risk, working at Bastion Lane Risk Advisory and Cedarstone Multi-Asset Group. On this team she owns risk oversight: volatility and drawdown monitoring, concentration limits when leaders cluster in correlated assets, and stress scenarios for sharp month-to-month reversals. Her distinctive strength is translating abstract risk metrics into clear, actionable limits that the committee can apply consistently each rebalance.

AI-generated fictional persona

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Digital Investment Team, continued (AI-generated, fictional)

Thomas Whitcombe

Macro and Cross-Asset Strategist

Focus: Macro and cross-asset context across bonds, gold, oil and credit

Certifications: CFA | **Experience:** 19 yrs

Education

BA, History and Economics, University of Oxford (2005)
MSc, Economics, London School of Economics (2007)

Experience

Rates and Commodities Analyst, Ashcombe Global Macro (2007-2015)
Senior Macro Strategist, Thornbury Cross-Asset Research (2015-2025)

Thomas Whitcombe has followed rates, commodities and credit markets since joining Ashcombe Global Macro after graduate study at the LSE. On this team he provides cross-asset context for Treasuries, gold, oil and high-yield credit, explaining what drives each ETF's behavior and how macro regimes affect relative trends. He does not override the model; instead he helps investors understand its positioning. His distinctive strength is historical perspective, drawing on past cycles to explain why defensive and risk assets rotate leadership.

AI-generated fictional persona

Sofia Lindqvist

Data Engineering Analyst

Focus: Market data engineering and backtest integrity

Certifications: None | **Experience:** 9 yrs

Education

MSc, Computer Science, KTH Royal Institute of Technology (2017)

Experience

Data Engineer, Northwind Data Systems (2017-2021)
Research Data Engineer, Silverpine Systematic Partners (2021-2025)

Sofia Lindqvist is a software engineer who built financial data pipelines at Northwind Data Systems before joining a systematic research team. On this team she owns market data quality and backtest integrity, including adjusted price handling for dividends and splits, month-end calendar logic, and safeguards against look-ahead and survivorship bias. Her distinctive strength is reproducibility: every backtest she maintains can be rerun from versioned code and data, giving the committee confidence that results reflect the rules as written.

AI-generated fictional persona

Marcus Delgado

Trading and Investor Communications Analyst

Focus: Trading, execution and investor communications

Certifications: CMT | **Experience:** 7 yrs

Education

BBA, Finance, University of Texas at Austin (2019)

Experience

Trading Associate, Lone Oak Brokerage Services (2019-2022)
Execution and Client Analyst, Cedarstone Multi-Asset Group (2022-2025)

Marcus Delgado began his career on an ETF trading desk at Lone Oak Brokerage Services before moving into a combined execution and client role. On this team he owns month-end rebalancing execution, monitoring spreads, liquidity and transaction costs, and he prepares the monthly investor update explaining which ETFs lead and why positions changed. His distinctive strength is clarity: he turns systematic rules and trade activity into concise, jargon-free communication that investors can follow.

AI-generated fictional persona

Important Disclosures

Hypothetical, in-sample performance. All performance shown is hypothetical and backtested. It does not represent actual trading or client accounts. The final strategies were selected because they had the highest total returns among 89 combinations tested on the same historical period, so their results benefit from hindsight (selection bias) and overstate what an investor could have expected in advance. No representation is made that any account will or is likely to achieve similar results.

Methodology. At each month end, each of the 7 ETFs (SPY, QQQ, TLT, IEF, GLD, USO, JNK) is scored by the % change of its signal (the close, or its N-day simple moving average) over a trailing lookback of D calendar days. Top 1 holds 100% of the top-ranked ETF; Top 3 mix holds 50%/30%/20% in the top three; Long top 3, short negatives adds an equal-weight short (100% in total) of every other ETF with a negative score. Holdings are bought at the month-end close and drift until the next month end. No trading costs are modeled. Returns use adjusted closing prices, which include dividends. Strategies are scored over a common window (2016-05-02 to 2026-10-28); the final year is year-to-date.

Costs and shorting. Results **exclude trading costs**, management fees and taxes; strategies that change holdings often would be affected most. The long/short variant ignores borrow costs and interest on short proceeds; short positions can lose more than their initial value.

Portfolio blends. SPY + strategy blends are rebalanced daily to fixed weights, without costs; this is a simplification of how a real portfolio would be managed.

Risk. Investing involves risk, including loss of principal. Ranking strategies can concentrate in one ETF, switch holdings frequently, lag trend reversals and underperform buy-and-hold for long periods. Oil (USO), high-yield credit (JNK), gold (GLD) and Treasury (TLT, IEF) ETFs carry their own risks, including sharp losses.

Data. Prices from Yahoo Finance via the yfinance library; not independently verified. The risk-free rate is the 13-week Treasury bill yield (^IRX).

AI-generated content and personas. The research, decisions, meeting script and slides were produced by AI agents and checked by code. The meeting is simulated. The investment team members are AI-generated fictional personas created by an AI system for illustration; they are not real people and their stated employers are fictional.

Not advice. This presentation is for informational purposes only, and is not investment advice, an offer, or a solicitation to buy or sell any security.